

BRAC University
Department of Computer Science and Engineering
CSE420:Compiler Design

Time: 35 min

Semester: Fall 25

Marks: 15

Name:	ID:	Section:
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1. Consider the following syntax directed translation. Construct the annotated parse tree for the given input string (4 marks), indicating the values of the attributes offset, type, and width at the relevant non terminal nodes. Additionally, show the contents of all symbol tables created during parsing (8 marks).

float [10][5] f ; record {int x ;} r ;

$$\begin{aligned}
 P &\rightarrow \{ \text{offset} = 0; \} D \\
 D &\rightarrow T \text{ id } ; \{ \text{top.put}(\text{id.lexeme}, T.\text{type}, \text{offset}); \text{offset} = \text{offset} + T.\text{width}; \} D_1 \\
 D &\rightarrow \epsilon \\
 T &\rightarrow B \{ t = B.\text{type}; w = B.\text{width}; \} C \{ T.\text{type} = C.\text{type}; T.\text{width} = C.\text{width} \} \\
 B &\rightarrow \text{int} \{ B.\text{type} = \text{integer}; B.\text{width} = 4; \} \\
 B &\rightarrow \text{float} \{ B.\text{type} = \text{float}; B.\text{width} = 8; \} \\
 C &\rightarrow \epsilon \{ C.\text{type} = t; C.\text{width} = w; \} \\
 C &\rightarrow [\text{num}] C_1 \{ C.\text{type} = \text{array}(\text{num.value}, C_1.\text{type}); C.\text{width} = \text{num.value} \times C_1.\text{width}; \} \\
 T &\rightarrow \text{record } \{ \{ \text{Env.push}(\text{top}); \text{top} = \text{new Env}(); \\
 &\quad \text{Stack.push}(\text{offset}); \text{offset} = 0; \} \\
 &\quad D \} \{ T.\text{type} = \text{record}(\text{top}); T.\text{width} = \text{offset}; \\
 &\quad \text{top} = \text{Env.pop}(); \text{offset} = \text{Stack.pop}(); \}
 \end{aligned}$$

2. After semantic analysis is complete, access modifiers must still be checked at runtime during field or method access. True or False? Justify your answer in 2–3 lines (3 marks)